PROTOTYPE DESCRIPTION

Smart Safety Companion

Overview

The Smart Safety Companion is a mobile application prototype developed using MIT App Inventor.

It is designed to enhance personal safety by detecting emergency situations through phone movement and automatically generating alerts with location and direction details.

Objective

The main objective of this prototype is to detect sudden or strong phone shaking and immediately alert emergency contacts with accurate location information.

User Interface Description

- The application interface contains the following components:
- Application title: Smart Safety Companion
- Start Monitoring button to activate monitoring
- Stop Monitoring button to stop monitoring
- Emergency contact number input
- Display of accelerometer values (X, Y, Z)
- Shake value display
- Location display (Latitude and Longitude)
- Emergency alert status message
- Direction display (North, East, South, West)

Purpose of the Prototype

- The main aim of this prototype is to improve personal safety.
- It performs the following functions:
- Monitors phone movement continuously
- Detects sudden or strong shaking
- Tracks the user's GPS location
- Identifies the direction of the device
- Generates emergency alert messages
- Stores emergency contact numbers

This app works as a safety assistant that reacts automatically in risky situations.

Application Title

The title "Smart Safety Companion" is displayed at the top of the screen.

Emergency Contact Section

Users can enter emergency contact numbers
Contacts are saved using TinyDB
Saved numbers remain even after closing the app
These contacts are used during emergencies

Start Monitoring Button

- Starts the monitoring process
- Activates accelerometer, location, and orientation sensors
- Changes the status to "Monitoring"
- Begins tracking phone movement

Stop Monitoring Button

- Stops all monitoring activities
- Turns off all sensors
- Resets values
- Changes the status to "Stopped"

Sensor and Status Display

- The app shows real-time data on the screen:
- Status – Stopped / Monitoring / Emergency Detected
- Accelerometer Values (X, Y, Z) – Phone movement data
- Shake Value – Calculated as |X| + |Y| + |Z|
- Location – Latitude and longitude
- Direction – North, East, South, or West
- Alert Message – Emergency alert with time, location, and direction

5. Block Programming Logic

Global Variables

- monitoring – Checks whether monitoring is ON or OFF
- cooldown – Prevents repeated alerts

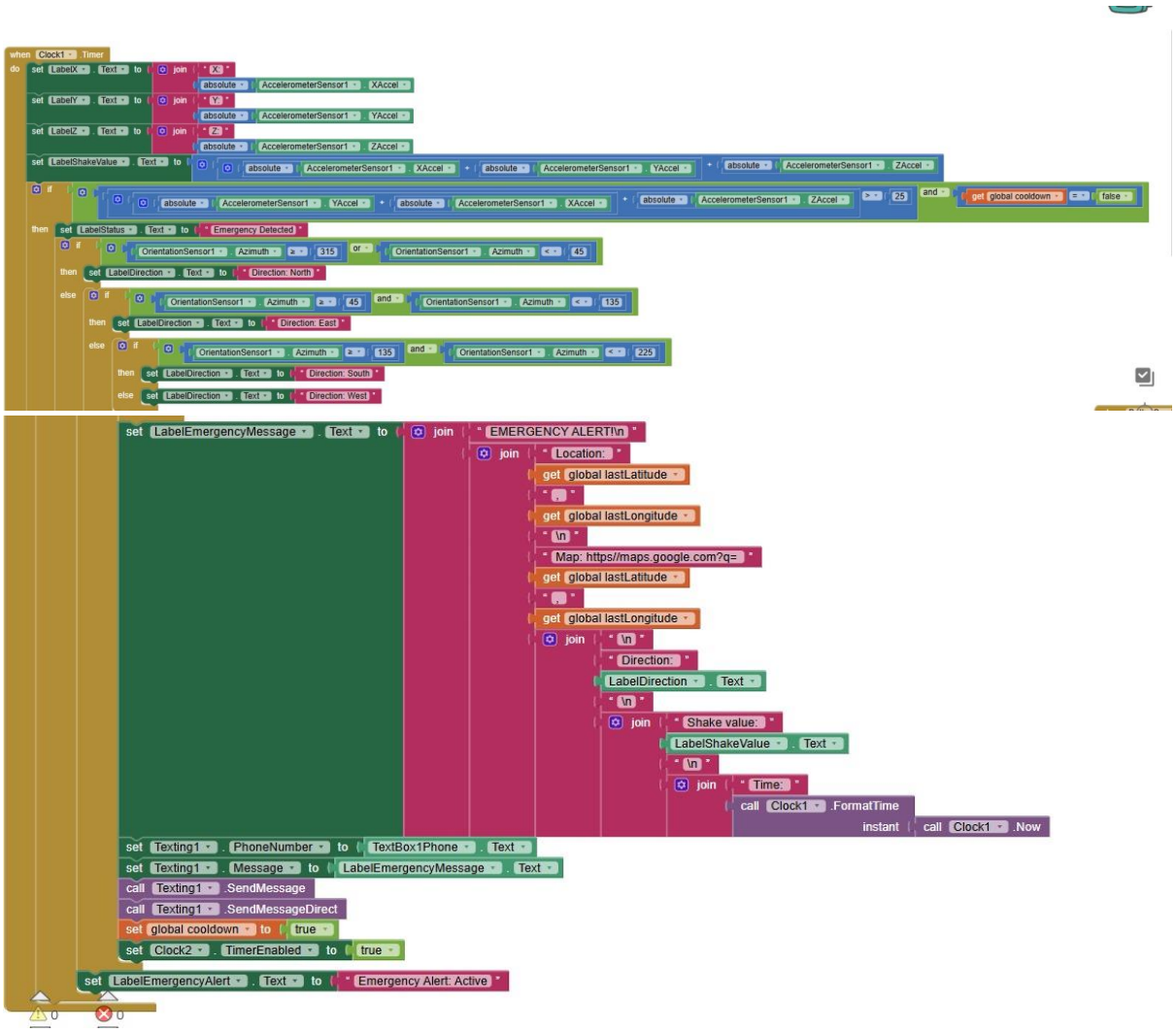
- shakeThreshold – Minimum shake value to detect emergency
- directionText – Stores direction information
- alertMessage – Stores alert message

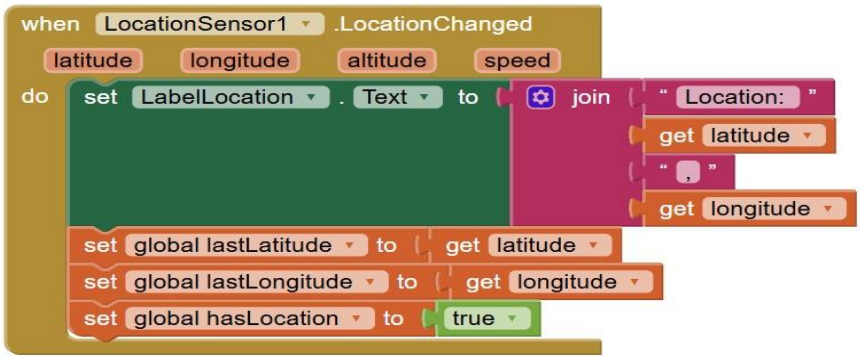
Working of Sensors

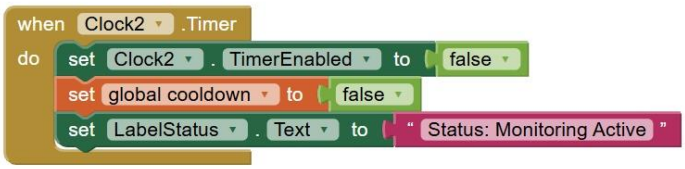
- Accelerometer checks phone movement
- Shake value is calculated using X, Y, Z values
- If shake value exceeds the threshold:
- Emergency is detected
- Location and direction are collected
- Alert message is generated
- Cooldown is activated
- Direction Detection

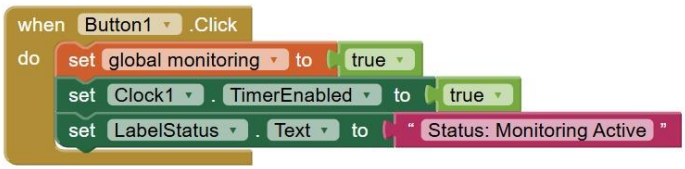
Using orientation sensor:

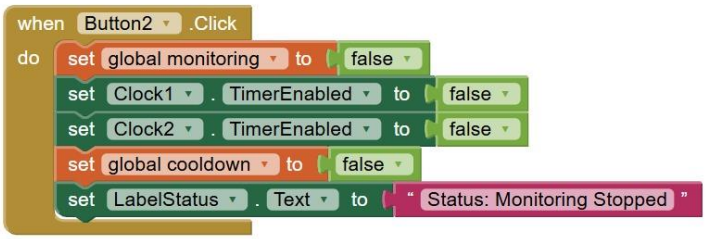
315° – 45° → North,45° – 135° → East,135° – 225° → South,225° – 315° → West











Overall Working

- User enters emergency contacts
- User starts monitoring
- App continuously checks phone movement
- If strong shaking is detected:
- Emergency alert is shown
- Location and direction are displayed
- User can stop monitoring at any time

Emergency Alert Generation

- When an emergency is detected
- Status changes to Emergency Detected

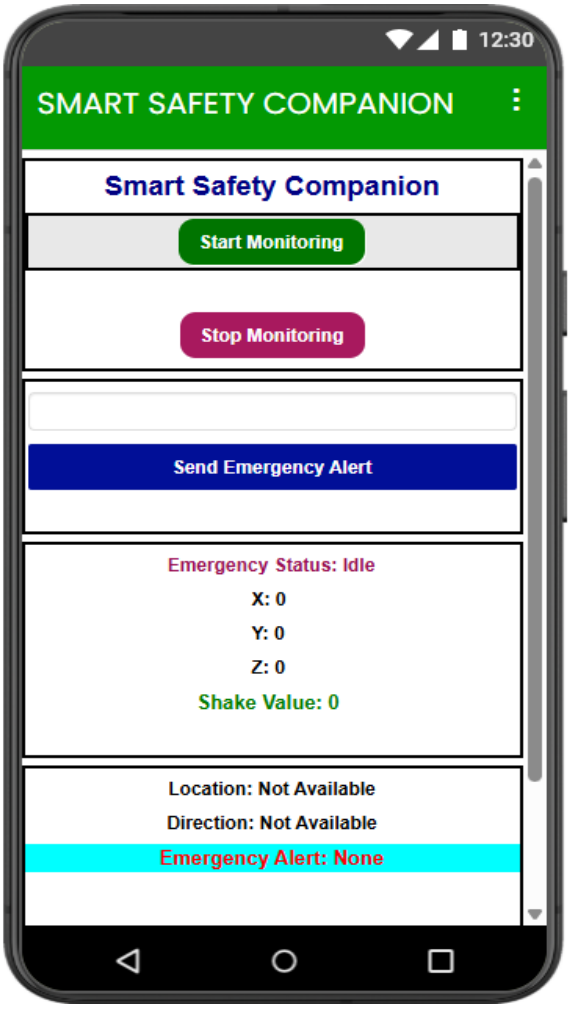
- Location, direction, shake value, and time are collected
- An emergency alert message is generated
- A Google Maps location link is created

SMS Alert and Cooldown

- The emergency message is sent automatically to the stored contact number using SMS.
- A cooldown mechanism is enabled to avoid repeated alerts.
- After cooldown, monitoring resumes automatically.

Output Display

- During emergency detection, the app displays:
- Emergency detected message
- Accelerometer value
- Shake value
- Current location
- Direction
- Emergency alert status as Active



Conclusion:

The Smart Safety Companion prototype successfully demonstrates the use of phone sensors to detect emergency situations.

By combining motion detection, GPS tracking, direction identification, and SMS alerts, the prototype provides an effective personal safety solution.